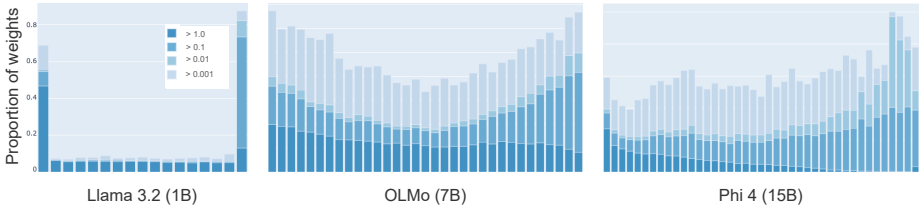


Figure 6: **Difference in probe weights distribution (of W_{in}) across layers** shows that the initial and last layers represent numbers in a systematically sparser fashion.

We hypothesize that this discrepancy can be caused by the models' computational and optimization mechanics. As cross-entropy loss is minimized by confident one-hot predictions, models can benefit from using more features to create higher dissimilarity (separability) of tokens. In contrast, internal representations are not subject to direct optimization pressure and might benefit from fewer but more informative features.

Nevertheless, we note that this trend is pertinent across different models to a different extent: OLMo 7B and Phi 15B learn more distributed features also across the intermediate layers — which, in turn, also leads to a better generalization of universal probes (Figure 5b), but still causing an outstanding drop in held-out accuracy compared to other layers. Finally, we note that the models' sparsity profile does not relate to reported performances on arithmetic tasks Kadlčík et al. (2025) and it also does not necessarily determine the accuracy of probes trained specifically for particular layer(s) (Figure 3).

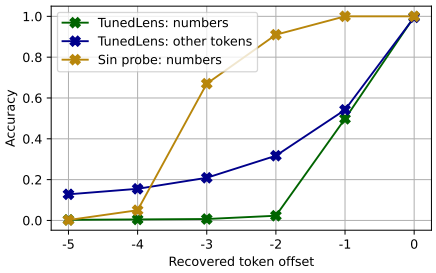


Figure 7: **Accuracy of previous tokens' recovery** for different probes trained on embeddings.

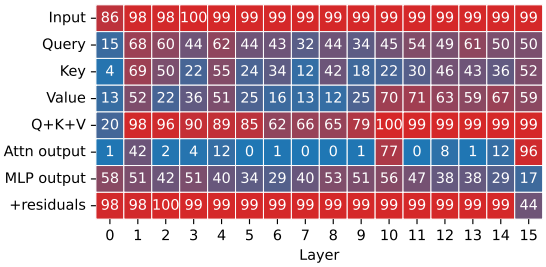


Figure 8: Accuracy of extracting input numbers from each component of Transformer block.

Finally, we explore the origin of the high consistency of representations across layers within the internal mechanism that each transformer layer implements. In Figure 8, we visualize the accuracy of probing input representations from each of the components present in transformer layers (Llama 3.2 1B for brevity). We find the accurate numeric representation is scattered across different components in the attention mechanism, with the attention output projection largely violating the sinusoidal representation, which is then, to a large extent, reconstructed in the subsequent fully-connected block. Nevertheless, the consistency of layers' output representation is primarily maintained by residual streams across layers.

To summarize, we find that the same probe can recover the numeric information for representations pooled from different layers of the models we study. Added to our previous observations in Sections 3.1 and 3.2, we can therefore stress that models learn input embeddings for numbers that are **strikingly similar**, regardless of which model they come from. These embeddings are processed into sinusoidal representations that are systematic and consistent regardless of context or hidden layer, with the source of this consistency originating primarily in the residual stream across layers.

4 MECHANISMS OF NUMBER MANIPULATION

We have established that models converge to sinusoidal number representations universally. This naturally begs the question of how those sinusoidal representations are used.

4.1 MULTI-TOKEN NUMBERS

Our first point of order concerns digits beyond the range of what can be represented in recent LLMs with a single token. To see whether LMs are able to represent even the values of multi-token numbers systematically and accurately, we probe the values of all parts of multi-token numbers from a single representation corresponding to the last numeric token. The accuracy of this experiment uncovers the extent to which the models implement a systematic algorithm of superposing multi-token numbers in a single representation.

For this experiment, we adapt the methodology utilizing natural-language contexts, with a substitution of original numeric values with values spanning two to six numeric tokens, i.e. in a numeric range between $10^3 - 10^{18}$. We use a subset of BBC data that was published after the knowledge cutoff or release date of all the models under consideration (Li et al., 2024).[5]

To be able to contextualize the magnitude of superposition and thus, disentangle a general model mechanism from the mechanism specific for numbers, we also probe representations of natural-language tokens using TunedLens (Belrose et al., 2025). TunedLens is an early-exit interpretability method which aims to explain the contents of intermediate representations by learning to translate them into logits. In practice, for each layer L_i, TunedLens involves distilling the computations done in the subsequent layers $L_{i+1}, \dots L_n$ into a simple affine transformation $\mathbf{W}_i$, such that applying this transformation $\mathbf{W}_i$ to a hidden state at L_i followed by the unembedding matrix closely matches the logits that the LLM would eventually produce. Here, to maximize comparability and align the number and distribution of target categories, we narrow down the space of probed tokens to 1000 tokens (matching the number probes) ranked as 2000–3000thmost-common tokens in our dataset.

Results for Llama 1B and different probing methods are presented in Figure 7. Results of sin probes show that **multi-token numbers are indeed systematically and highly accurately superposed in the latest numeric representation** – reaching an accuracy of 99% for the immediately-preceding number piece (offset -1). However, the accuracy and/or systematicity of the superposition mechanism quickly drops for numbers longer than three tokens (i.e. $\geq 10^9$), reaching close to zero for the 5th preceding token.

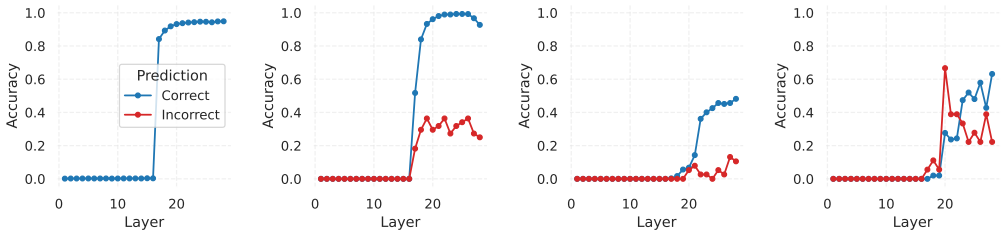
The results of natural-language token probing using TunedLens reveal that superpositioning representations of previous tokens is *not* a mechanism specific only to numbers – albeit this mechanism seems more prevalent and accurate for numeric tokens. Aiming to maximise comparability, we also report a comparison of the previous-token recovery using TunedLens for *numeric* tokens — in exactly the same configuration as for the natural-language tokens. This comparison shows that TunedLens is highly inefficient for recovering numeric tokens compared to sinusoidal probes.

4.2 ERROR TRACKING

Establishing that LLMs use universal, systematic and interchangeable representations of numbers, we aim to explore whether we can build upon this knowledge in tracing the origins of models' outputs in arithmetic reasoning. First, we assess the predictive power of models' internal representations towards true results in prompts requiring addition, subtraction, multiplication and division.

Results of predicting the Llama 3.2 3B's expected outputs for different operations are displayed in Figure 9. We can see a large distinction between the cases where the model predicts a correct and incorrect response – suggesting that the **extent to which the model maintains the sinusoidal representation may determine the accuracy of the output**. We observe that for addition and subtraction, probes can, with close to 100% accuracy, identify the retrieved result already from the model internals. Based on qualitative assessments of output embeddings (subsection B.1), we hypothesise that lower reliability in other operations (multiplication and division) may be caused by

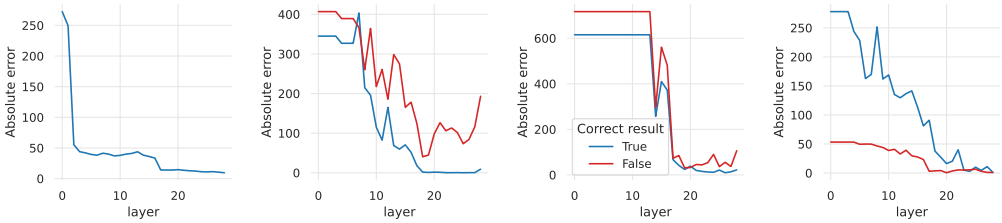
[5]Retrieved from `RealTimeData/bbc_news_alltime`



(a) Addition (b) Subtraction (c) Multiplication (d) Division

Figure 9: Probing accuracy in predicting models' outputs for different prompted operations, for cases where the model predicts a correct (blue) and incorrect response (red).

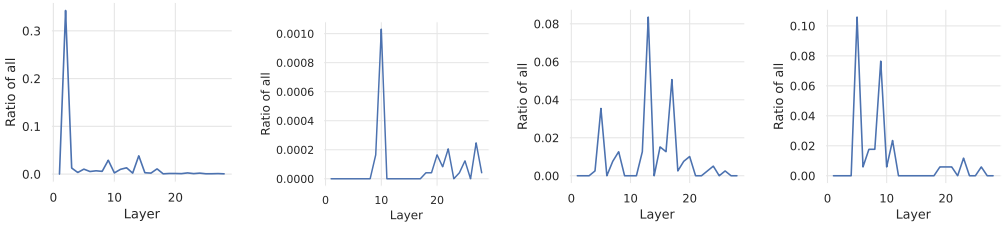
the models' divergence from sinusoidal representation employed by the input/output embeddings, accompanied also by lower accuracy of models overall (Table 2).



(a) Addition (b) Subtraction (c) Multiplication (d) Division

Figure 10: **Absolute error per layer:** Absolute error of numeric values probed from different layers.

The results in Figure 10 show that across all operations, models tend to incrementally reduce the error towards the true answer value, *gradually* refining across layers. These results also suggest that there are particular layers responsible for an increase of errors in the model's internal computation.



(a) Addition (b) Subtraction (c) Multiplication (d) Division

Figure 11: **Error aggregation per layer:** Relative ratio of cases where each of the layers in Llama 3.2 3B *breaks* the correct result from the previous layer. We find that removing the three layers with the highest error aggregation in division (layers 5, 9 and 11) leads to 27–64% error reduction.

Figure 11 corroborates this hypothesis — showing that particular layers break the correct result probed from the previous layer in large proportions of all predictions. For instance, in division, the 5th layer is responsible for breaking the correct result recoverable from the previous layer in over 10% of all cases. Our further analyses, overviewed in Table 2, show that according to the probed representations, the models often achieve a correct result *internally*, even though it does *not* surface: this is the case of 56.8% of all surfaced (prediction) errors in subtraction, 26.3% in multiplication, and as much as 94.4% of all errors in division.

We further support the claimed responsibility of specific layers on the resulting accuracy by simply removing the suspected layers from the model. Such a naive alternation becomes viable with the un-